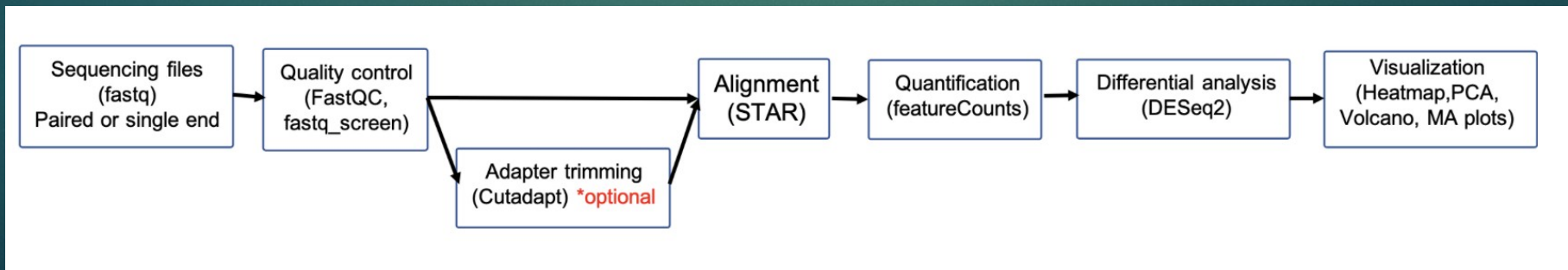


RNA-Seq best practices

JAMES ROUSE

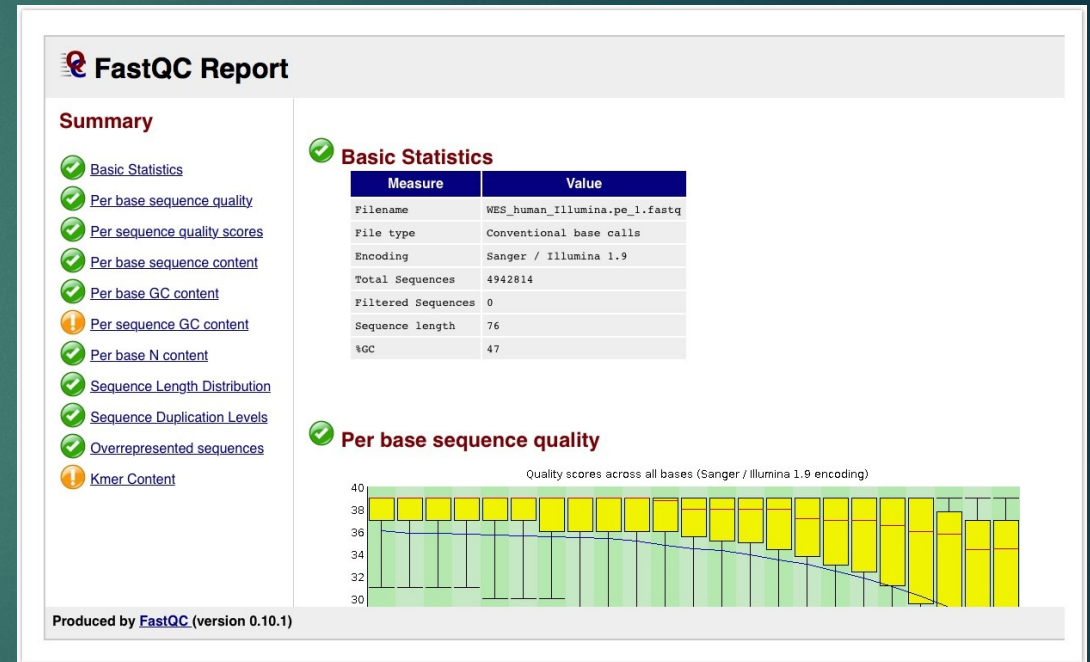
Introduction to RNA-Seq best practices

- ▶ **Quality Control:** Use FastQC for quality checks and Trimmomatic or Cutadapt for trimming low-quality bases and adapters.
- ▶ **Alignment/Quantification:** Use STAR or HISAT2 for alignment, or Salmon/Kallisto for faster quantification without alignment. Htseq-count or featureCounts used for quantification with alignment.
- ▶ **Differential Expression:** Use DESeq2, edgeR, or limma for normalization and differential expression, accounting for technical biases.
- ▶ Some tools perform better than others, especially when dealing with certain types of data



Best practices during quality control

- ▶ **Purpose:** Assess the quality of the reads and ensure that the sequencing data is of high quality and reliable for downstream analysis.
- ▶ **FastQC:** Identifies potential issues such as poor read quality, adapter contamination, and sequence biases.
- ▶ **Fastq_screen:** Aligns reads to multiple reference genomes/databases. Good way to detect contamination.
- ▶ Resequencing often recommended if the results of the FastQC report are poor.



Best practices during trimming

- ▶ **Adapter trimming:** Adapter trimming is optional, but often recommended. Some alignment tools can handle adapters. Trimmomatic and CutAdapt are two common tools used for trimming.
- ▶ [Williams et al.](#) argues that quality-based trimming may negatively impact the downstream analysis. They note that differences come from the short reads that result from trimming and suggest filtering reads by length after trimming.

Best practices during alignment

- ▶ **STAR:** STAR is an alignment tool known for its accuracy. STAR is also more robust to variant detections. Allows for discovery of novel splice junctions
- ▶ **HISAT2:** Memory efficient compared to STAR, but sometimes slightly less accurate.
- ▶ **Kallisto:** Pseudoalignment, maps reads to a set of transcripts rather than to a reference genome. Faster and more memory efficient than traditional alignment, but sometimes less accurate.
- ▶ Mapping percentage: 90% - ideal, 80% - good, 70% - acceptable, 60% - borderline, <50% - resequence

Evaluation of 7 different alignment tools

- ▶ [Schaarschmidt et al.](#) - bwa, CLC Genomics Workbench, HISAT2, kallisto, RSEM, salmon and STAR were all used for alignment in model plant *Arabidopsis thaliana*
- ▶ All tools had similar mapping rates and there was high overlap of mappers.
- ▶ STAR had the highest fraction of mapped reads, likely due to higher tolerance for soft-clipped and mismatched bases.
- ▶ Generally, all tools were comparable to each other.

Comparison of 7 different alignment tools

- ▶ [Musich et al.](#) – Comparative analysis of different alignment tools on *Erysiphe necator*. Runtime, alignment percentage, and transcriptome coverage measured.
- ▶ **TopHat2** showed the lowest alignment rates and the slowest runtime.
- ▶ **Bowtie2 (Local mode)** and **BWA** achieved the highest alignment rates (~87%).
- ▶ **HISAT2** was the fastest aligner, with a runtime of 8.28 ms per read.
- ▶ **STAR** and **HISAT2** performed better for mapping larger transcripts.
- ▶ **TopHat2** had the highest number of unmapped genes (1,570), while **BWA** had the lowest (106).

De novo alignment

- ▶ De novo genome assembly reconstructs a genome from sequencing reads without a reference genome by aligning overlapping sequences.
- ▶ [Jeon et al.](#) - For long reads (e.g., PacBio, ONT), tools like Canu and Flye are optimal for high contiguity and accuracy.
- ▶ For short reads (e.g., Illumina), SPAdes, Trinity, and ABySS are highly effective for assembling small to medium-sized genomes

Alignment-free tools/pseudoaligners

- ▶ Pseudo-aligners like Kallisto don't align the reads to a reference base by base. Rather, they determine which transcripts a read is compatible with.
- ▶ This technique is fast and effective for transcript quantification
- ▶ [Wu et al.](#) - Alignment-free RNA-seq pipelines struggle with accurate quantification of small and lowly-expressed RNAs, particularly tRNAs

Potential strength of alignment-free tools

- ▶ [Zheng et al.](#) - Dependent on type of data and goal of analysis. During analysis to quantify lncRNA in TCGA, salmon and kallisto outperformed STAR.
- ▶ Salmon and kallisto detected more lncRNAs and stronger concordance with the ground truth.

Best practices during quantification

- ▶ HTSeq-count is slower and more memory-intensive and discards multi-mapping reads by default.
- ▶ featureCounts is faster, more memory-efficient, and handles multi-mapping reads. It is better suited for large-scale, high-throughput workflows.
- ▶ [Corchete et al.](#) - Generally, pipelines that use HTSeq-count showed better accuracy when compared to featureCounts in identifying gene counts.
- ▶ [Jin et al.](#) TPM is better than raw counts because it accounts for both gene length and sequencing depth.
- ▶ TPM demonstrated better linearity compared to raw counts, as its expression values more closely matched the true mixture proportions and showed stronger linear correlations with the actual gene abundance

Quantification of transcripts

- ▶ Transcript-level analysis allows for the detection of specific isoforms that may be differentially expressed
- ▶ **RSEM** is particularly useful for identifying isoforms, as it quantifies transcript-level expression, enabling investigation of alternative splicing and isoform-specific expression changes.

TPM is better than raw counts, but should it always be used?

- ▶ **TPM normalizes for sequencing depth and gene length** but does not account for **RNA composition bias**, meaning it can be skewed when a few genes are highly expressed in one sample.
- ▶ **DESeq2 accounts for RNA composition bias**, making it more reliable for comparing gene expression between samples with different RNA compositions.

Best practices during differential gene expression

- ▶ **edgeR** tends to find more DEGs and works efficiently with smaller sample sizes or when you need more statistical power, but at the cost of potentially more false positives.
- ▶ **DESeq2** produces fewer but more reliable DEGs, making it a good choice for experiments where confidence in results is critical, especially in larger datasets with more biological replicates.

Normalization methods in quantification tools

- ▶ TMM: Calculates a scaling factor by trimming extreme log-fold changes between samples; adjusts for compositional differences and is robust to outlier genes
- ▶ RLE: Compares each gene's expression to the median across all samples; calculates scaling factors from median log ratios; assumes most genes are similarly expressed
- ▶ MRN: Uses the median of all replicates in a condition as the reference; provides a condition-specific normalization factor; aims for consistency and stability across replicates.

Which normalization method should be used and when?

- ▶ TMM: Use for large compositional differences between samples; suitable for both within-sample and between-sample comparisons; ideal for complex designs needing robustness to outliers.
- ▶ RLE: Use when most genes are similarly expressed; suitable for between-sample comparisons; effective for simple experimental designs.
- ▶ MRN: Prefer for complex designs with multiple conditions and replicates; ensures condition-specific normalization and stability across replicates.
- ▶ [Maza et al.](#) – MRN performs better on simulated datasets in experimental designs with two conditions and multiple replicates.

DeSeq2 and edgeR may exaggerate false positives

- ▶ [Li et al.](#) - DESeq2 and edgeR showed large discrepancies in the DEGs they identified across 13 population-level RNA-seq datasets, with false positives often inflated due to poor FDR control, especially in large-sample-size datasets.
- ▶ The Wilcoxon rank-sum test and limma consistently outperformed DESeq2, edgeR, by better controlling FDR and maintaining statistical power.
- ▶ Permutation analysis showed inflated FDRs. The methods' performance issues were linked to model violations and outliers.

Best practices for pathway analysis

- ▶ GSEA (Gene Set Enrichment Analysis): Uses a ranked gene list to identify whether predefined gene sets are enriched at the top or bottom of the ranked list. It provides detailed outputs like normalized enrichment scores (NES) and enrichment plots.
- ▶ Enrichr: A fast and user-friendly tool for analyzing a predefined list of significant genes, comparing it to a wide range of gene set databases (e.g., GO, KEGG, Reactome).

Best practices for visualization

- ▶ **Sletuh:** Used in combination with Kallisto. Not well maintained, but provides easy visualization if you have pseudoaligned with Kallisto. Produces PCA plots and can give you the PCA loadings.
- ▶ **DeSeq2 and edgeR:** Tools have built in visualization methods for RNA-Seq. Often accessible in R.
- ▶ **R:ggplot2 & Bioconductor packages:** Essential for high-quality plots and specialized bioinformatics visualizations (e.g., DESeq2, edgeR).

Troubleshooting using visualization

- ▶ PCA Loadings Analysis:
 - ▶ Examine PCA loadings to identify the most important genes contributing to principal components. If housekeeping genes or lncRNAs dominate the loadings, differences may be superficial rather than biologically meaningful.
 - ▶ Identifying such patterns can help in deciding if certain samples or genes are causing misleading variation.
- ▶ Troubleshooting with RNA-Seq Visualizations:
 - ▶ Use PCA plots to assess clustering; poorly clustered replicates could indicate technical issues.
 - ▶ MA and volcano plots help identify problematic genes with unexpected behavior across conditions. For example, a gene well known to be upregulated in breast cancer is downregulated in breast cancer samples.

Gene Fusion

- ▶ **Gene Fusion:** Gene fusions occur when parts of two different genes are joined together. These fused genes are often implicated in cancer.
- ▶ Tools to Detect Gene Fusion:
 - ▶ **STAR-Fusion:** A popular tool for detecting gene fusions from RNA-seq data using the STAR aligner.
 - ▶ **Kallisto:** Has an option for gene fusion detection in addition to its main function of quantifying transcript abundances.

Other quality control tools

- ▶ RSeQC: Performed after read alignment, before downstream analysis. Purpose: Provides metrics like read distribution, gene body coverage, GC content, helping detect RNA degradation and technical biases.
- ▶ **MultiQC**: Aggregates QC reports to identify trends, batch effects, and outliers, providing a summary across all samples

Summary

- ▶ **Quality Control:** **FastQC** and **Fastq_screen** are used for assessing read quality and contamination. Trimming low-quality reads may help but can negatively affect analysis.
- ▶ **Alignment:** **STAR** is used for high precision and novel splice detection, while **Kallisto** is faster and efficient for transcript-level quantification. Alignment-based tools offer accuracy but are computationally intensive, whereas alignment-free tools are faster but have limitations in feature detection.
- ▶ **Quantification:** **featureCounts** is faster, while **HTSeq-count** is more accurate. TPM is preferred for comparing expression across samples because it accounts for gene length and sequencing depth.
- ▶ **Differential Gene Expression:** **DESeq2** provides robust, accurate analysis of differential expression, whereas **edgeR** identifies more DEGs but has a higher risk of false positives.
- ▶ **Pathway Analysis:** **GSEA** is used for ranked gene set enrichment, while **Enrichr** provides a fast, user-friendly tool for analyzing predefined gene lists.